

Lab 9: Inheritance Genetics — Genetics Problems

BIOL-1

Name: _____ Date: _____

Overview

In this lab, you will practice solving genetics problems using Punnett squares. You will work through simple crosses, blood type problems, sex-linked traits, and a pedigree.

Materials

- This worksheet
- Scratch paper for Punnett squares

Part 1: Monohybrid Crosses

In pea plants, Tall (T) is dominant to Short (t).

1. Cross a heterozygous tall plant (Tt) with a short plant (tt). Draw the Punnett square below.

t t

T

t

Genotypic ratio: ____

Phenotypic ratio: ____

2. Cross two heterozygous tall plants (Tt × Tt). Draw the Punnett square.

T t

T

t

Genotypic ratio: ____

Phenotypic ratio: ____

Part 2: Incomplete Dominance

In snapdragons: Red (RR) $\times$ White (R'R') = Pink (RR')

3. Cross two pink snapdragons (RR' $\times$ RR').

Phenotypic ratio: **Red** : Pink : ____ White

4. How is incomplete dominance different from complete dominance?

Part 3: Blood Types (Codominance + Multiple Alleles)

5. Fill in the genotypes for each blood type:

Blood Type Possible Genotypes

Type A

Type B

Type AB

Type O

6. A mother is Type A (I^Ai) and a father is Type B (I^Bi). Draw the Punnett square.

I^B i

I^A

i

7. What blood types are possible for their children?

8. Can they have a child with Type O blood? Explain.

Part 4: Sex-Linked Traits

Color blindness is X-linked recessive. Normal vision = X^B. Color blind = X^b.

9. A carrier mother ($X^B X^b$) and a normal father ($X^B Y$) have children. Draw the Punnett square.

$X^B Y$

X^B

X^b

10. What fraction of their sons will be colorblind? ____

11. What fraction of their daughters will be carriers? ____

12. Why are X-linked recessive traits more common in males?

Part 5: Pedigree Analysis

Look at this pedigree description: Two unaffected parents have an affected son and an unaffected daughter. The unaffected daughter later has an affected son of her own.

13. Is this trait dominant or recessive? How do you know?

14. Is this trait autosomal or X-linked? Explain your reasoning.

Part 6: Polygenic Traits

15. What is a polygenic trait? Give one example.

16. How is the inheritance of human height different from the inheritance of tall/short pea plants?

17. True or False: If you know someone's genotype, you always know their exact phenotype. Explain.
